# Lecture Notes: Chapter 5 and Chapter 6

Sections 5.1, 5.2, 5.3, 5.4, 6.1

Spring, 2020

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## Contents

## Section 5.1 - Approximating Areas Under Curves

The derivative of a function is associated with rates of change and slopes of tangent lines. We also know that antiderivatives (or indefinite integrals) reverse the derivative operation. This raises the question: What is the geometric meaning of the integral? To answer this question, let's start with an example. Imagine a car traveling at a constant velocity of 60 $mi/hr$ along a straight highway over a two-hour period. The graph of the velocity function $v = 60$ on the interval $0 \leq t \leq 2$ is a horizontal line (Figure 1). The displacement of the car between $t = 0$ and $t = 2$ is found by a familiar formula:

$$\text{displacement} = \text{rate} \times \text{time} = 60 \ \ mi/hr \times 2 \ \ hr = 120 \ \ mi.$$

This product is the area of the rectangle formed by the velocity curve and the $t$-axis between $t = 0$ and $t = 2$ (Figure 2). In the case of constant positive velocity, we see that the area between the velocity curve and the $t$-axis is the displacement of the moving object.

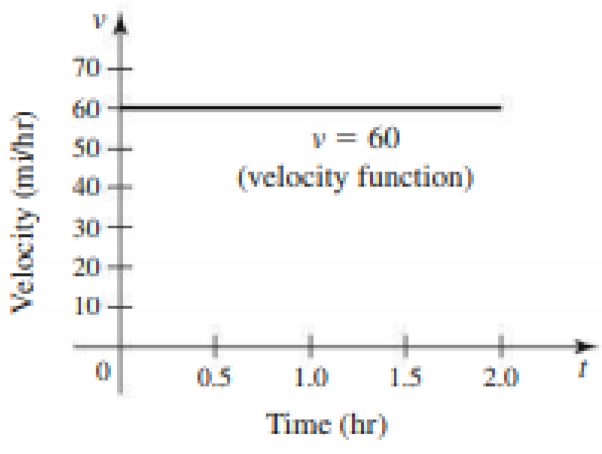


Figure 1: The graph of the velocity function $v = 60$ on the interval $0 \leq t \leq 2$

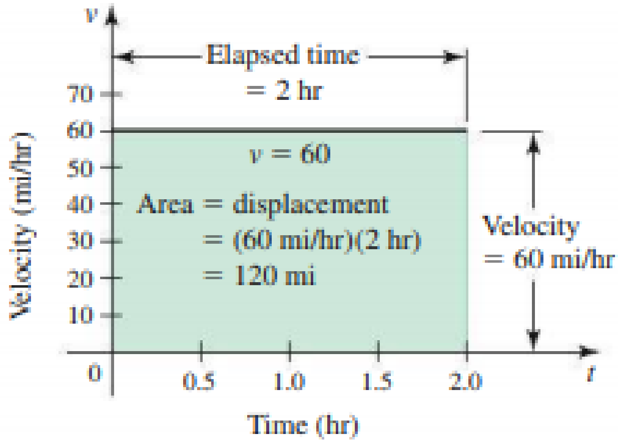


Figure 2: The area formed by the velocity curve and the $t$-axis between $t = 0$ and $t = 2$

Notice that the indefinite integral of $v$ is given by $\displaystyle\int v \ \ dt = \int 60 \ \ dt = 60t + c$. So, as the derivative of a position yields velocity, let's call this function $p(t)$. Now, we may observe that, at time $t = 0$ $hr$, the position is given by $p(0) = c$ $mi$, and at time $t = 2$ $hr$, the position is given by $p(2) = (60(2) + c)$ $mi = (120 + c)$ $mi$. So, regardless of the value of $c$, we can see that

$$p(2) - p(0) = \Big((120 + c) - c\Big) \ \ mi = 120 \ \ mi = \ \text{displacement}.$$

So, in the case of a constant velocity function $v$, we see that the integral of $v$ gives us some information about the net distance we've traveled (i.e. the displacement).

However, because objects do not necessarily move at a constant velocity, we must first extend these ideas to positive velocities that change over an interval of time before we go any further. One strategy is to divide the time interval into many subintervals and approximate the velocity on each subinterval with a constant velocity. Then the displacements on each subinterval are calculated and summed. This strategy produces only an approximation to the displacement; however, this approximation generally improves as the number of subintervals increases. Basically, just picture something along the lines of the process illustrated in Figure 3 below:

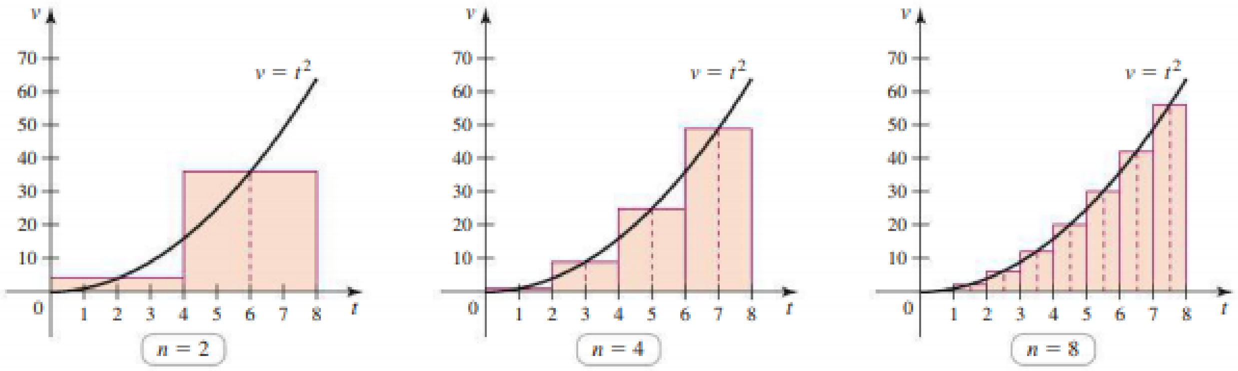


Figure 3: The midpoint of each subinterval is used to approximate the velocity over that subinterval.

Now, let's step back a moment. Of course, we wouldn't spend much time investigating areas under curves if the idea applied only to computing displacements from velocity curves. However, the problem of finding areas under curves arises frequently and turns out to be immensely important, as you will see throughout the remainder of the course (and in further calculus classes, if you choose to take any). For this reason, we now develop a systematic method for approximating areas under curves.

Consider a function $f$ that is continuous and nonnegative on an interval $[a,b]$. The goal is to approximate the area of the region R bounded by the graph of $f$ and the $x$-axis from $x = a$ to $x = b$ (see Figure 4).

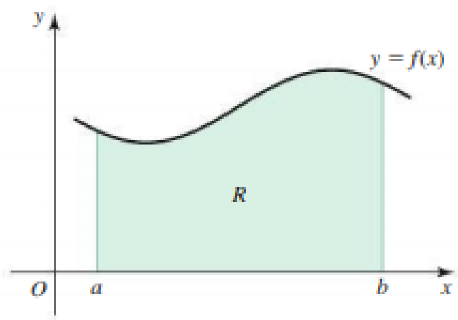


Figure 4: The region R bounded by the graph of $f$ and the $x$-axis from $x = a$ to $x = b$

We begin by dividing the interval $[a,b]$ into $n$ subintervals of equal length,

$$[x_0, x_1], \ [x_1, x_2], \ \ldots, \ [x_{n-1}, x_n]$$

where $a = x_0$ and $b = x_n$. The length of each subinterval, denoted $\Delta x$, is found by dividing the length of the entire interval by $n$:

$$\Delta x = \frac{b-a}{n}.$$

We can use this partition of $[a,b]$ into subintervals to approximate the area under a curve (as shown in Figure 4) by the method shown in Figure 3. This leads us to the idea of a Riemann sum, which we will now define.

**Definition 1** (Riemann Sum).
Suppose $f$ is defined on a closed interval $[a,b]$, which is divided into $n$ subintervals of equal length $\Delta x$. If $x_k^*$ is any point in the $k^{th}$ subinterval $[x_{k-1}, x_k]$, for $k = 1, 2, \ldots, n$, then

$$f(x_1^*)\Delta x + f(x_2^*)\Delta x + \ldots + f(x_n^*)\Delta x$$

is called a Riemann sum for $f$ on $[a,b]$. This sum is called

- a left Riemann sum if $x_k^*$ is the left endpoint of $[x_{k-1}, x_k]$ i.e. if $x_k^* = x_{k-1}$
- a right Riemann sum if $x_k^*$ is the right endpoint of $[x_{k-1}, x_k]$ i.e. if $x_k^* = x_k$
- a midpoint Riemann sum if $x_k^*$ is the midpoint of $[x_{k-1}, x_k]$ i.e. if $x_k^* = \dfrac{x_k - x_{k-1}}{2}$.

**Example 1.** Let R be the region bounded by the graph of $f(x) = \sin(x)$ and the $x$-axis between $x = 0$ and $x = \pi/2$. Approximate the area of R using a left Riemann sum with $n = 6$ subintervals. Then, approximate the area of R using a right Riemann sum with $n = 6$ subintervals.

Dividing the interval $[a,b] = [0, \pi/2]$ into $n = 6$ subintervals means the length of each subinterval is

$$\Delta x = \frac{b-a}{n} = \frac{\pi/2 - 0}{6} = \frac{\pi}{12}.$$

So, to find the left Riemann sum, we set $x_1^*, x_2^*, \ldots, x_6^*$ equal to the left endpoints of the six subintervals. The heights of the rectangles are $f(x_k^*)$ for $k = 1, \ldots, 6$. So, the left Riemann sum is

$$\begin{aligned} f(x_1^*)\Delta x + f(x_2^*)\Delta x + \ldots + f(x_6^*)\Delta x &= sin\left(\frac{0\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{2\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{3\pi}{12}\right)\frac{\pi}{12} \\ &\quad + sin\left(\frac{4\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{5\pi}{12}\right)\frac{\pi}{12} \\ &\approx 0.863. \end{aligned}$$

Similarly, in a right Riemann sum, the right endpoints are used for $x_1^*, x_2^*, \ldots, x_6^*$, and the heights of the rectangles are $f(x_k^*)$, for $k = 1, \ldots, 6$. So, the right Riemann sum is

$$\begin{aligned} f(x_1^*)\Delta x + f(x_2^*)\Delta x + \ldots + f(x_6^*)\Delta x &= sin\left(\frac{\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{2\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{3\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{4\pi}{12}\right)\frac{\pi}{12} \\ &\quad + sin\left(\frac{5\pi}{12}\right)\frac{\pi}{12} + sin\left(\frac{6\pi}{12}\right)\frac{\pi}{12} \\ &\approx 1.125. \end{aligned}$$

As you've probably noticed, working with Riemann sums is cumbersome with large numbers of subintervals. Therefore, we pause for a moment to introduce some notation that simplifies our work. Sigma (or summation) notation is used to express sums in a compact way. For example, the sum $1+2+3+\ldots+10$ is represented in sigma notation as

$$\sum_{k=1}^{10} k = 1+2+3+\ldots+10.$$

Here is how the notation works: The symbol $\sum$ (sigma, the Greek capital S) stands for sum. The index $k$ takes on all integer values from the lower limit ($k=1$, in the above example) to the upper limit ($k=10$ in the above example). The expression that immediately follows $\sum$ (called the summand) is evaluated for each value of $k$, and the resulting values are summed. Here are some examples:

$$\sum_{k=1}^{99} k = 1+2+3+\ldots+98+99, \qquad \sum_{k=1}^{3} k^2 = 1^2+2^2+3^2, \qquad \sum_{k=1}^{3}\left(2k^2-1\right) = \left(2(1)^2-1\right)+\left(2(2)^2-1\right)+\left(2(3)^2-1\right)$$

The index in a sum is a dummy variable. It is internal to the sum, so it does not matter what symbol you choose as an index. For example,

$$\sum_{k=1}^{999} k = \sum_{n=1}^{999} n = \sum_{t=1}^{999} t$$

Of course, since we can factor a constant multiple out of a sum, we can factor a constant multiple out of a sum written in sigma notation. So, statements like

$$\sum_{k=1}^{43} 9(k+2) = 9\sum_{k=1}^{43}(k+2)$$

are totally fine and mathematically valid. Similarly, we can break sums up along indices

$$\sum_{k=1}^{43} 9(k+2) = \left(\sum_{k=1}^{16} 9(k+2)\right) + \left(\sum_{k=17}^{43} 9(k+2)\right)$$

or along terms of the sum

$$\sum_{k=1}^{11}\left(k+k^2\right) = \left(\sum_{k=1}^{11} k\right) + \left(\sum_{k=1}^{11} k^2\right).$$

Here are some useful facts about certain sums:

**Theorem 1** (Sums of Powers of Integers).
*Let $c$ be a constant and $n$ a positive integer. Then, we have the following:*

$$\sum_{k=1}^{n} c = cn, \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}.$$

Now, we can redefine Riemann sums in terms of this useful notation.

**Definition 2** (Left, Right, and Midpoint Riemann Sums in Sigma Notation).
Suppose $f$ is defined on a closed interval $[a,b]$, which is divided into $n$ subintervals of equal length $\Delta x$. If $x_k^*$ is any point in the $k^{th}$ subinterval $[x_{k-1}, x_k]$, for $k = 1, 2, \ldots, n$, then the Riemann sum of $f$ on $[a,b]$ is $\sum_{k=1}^{n} f\left(x_k^*\right)\Delta x$. Three cases arise in practice:

- Left Riemann Sum: $\sum_{k=1}^{n} f\Big(a+(k-1)\Delta x\Big)\Delta x$, i.e. if $x_k^* = x_{k-1}$
- Right Riemann Sum: $\sum_{k=1}^{n} f\Big(a+k\Delta x\Big)\Delta x$, i.e. if $x_k^* = x_k$
- Midpoint Riemann Sum: $\sum_{k=1}^{n} f\Big(a+(k-1/2)\Delta x\Big)\Delta x$, i.e. if $x_k^* = \dfrac{x_k - x_{k-1}}{2}$.

**Example 2.** Evaluate the left Riemann sum for $f(x) = x^3 + 1$ between $a = 0$ and $b = 2$ using $n = 50$ subintervals.

With n = 50, the length of each subinterval is $\Delta x = \dfrac{2-0}{50} = 0.04$. The value $x_k^*$ for the left Riemann sum is $x_k^* = a + (k-1)\Delta x = 0 + 0.04(k-1) = 0.04k - 0.04$. Thus,

$$\begin{aligned} f\left(x_k^*\right) = f\Big(a+(k-1)\Delta x\Big) &= f\Big(0.04k - 0.04\Big) \\ &= \Big(0.04k-0.04\Big)^3 + 1 \\ &= 0.04^3k^3 - 3(0.04)^3k^2 + 3(0.04)^3k - 0.04^3 + 1 \end{aligned}$$

So, the Left Riemann sum is

$$\begin{aligned} \sum_{k=1}^{50} f\Big(a+(k-1)\Delta x\Big)\Delta x &= \sum_{k=1}^{50}\Big(0.04^3k^3 - 3(0.04)^3k^2 + 3(0.04)^3k - 0.04^3 + 1\Big)0.04 \\ &= 0.04\sum_{k=1}^{50}\Big(0.04^3k^3 - 3(0.04)^3k^2 + 3(0.04)^3k - 0.04^3 + 1\Big) \\ &= 0.04\left(0.04^3\left[\sum_{k=1}^{50}k^3\right] - 3(0.04)^3\left[\sum_{k=1}^{50}k^2\right] + 3(0.04)^3\left[\sum_{k=1}^{50}k\right] + \left[\sum_{k=1}^{50}(1-0.04^3)\right]\right) \\ &= 0.04\left(0.04^3\left(\frac{(50)^2(51)^2}{4}\right) - 3(0.04)^3\left(\frac{50(51)(101)}{6}\right) + 3(0.04)^3\left(\frac{50(51)}{2}\right) + (1-0.04^3)(50)\right) \\ &= 5.8416. \end{aligned}$$

## Section 5.2 - Definite Integrals

We introduced Riemann sums in Section 5.1 as a way to approximate the area of a region bounded by a curve $y = f(x)$ and the $x$-axis on an interval $[a,b]$. In that discussion, we assumed $f$ to be nonnegative on the interval. Our next task is to discover the geometric meaning of Riemann sums when $f$ is negative on some or all of $[a,b]$. Once this matter is settled, we proceed to the main event of this section, which is to define the definite integral. With definite integrals, the approximations given by Riemann sums become exact.

How do we interpret Riemann sums when f is negative at some or all points of $[a,b]$? The answer follows directly from the Riemann sum definition. That is, on intervals where $f(x) < 0$, Riemann sums approximate the *negative* of the area of the region bounded by the curve. In the more general case that f is positive on only part of $[a,b]$, we get *positive* contributions to the sum where $f$ is positive and *negative* contributions to the sum where $f$ is negative. In this case, Riemann sums approximate the area of the regions that lie *above* the $x$-axis *minus* the area of the regions that lie *below* the $x$-axis. This leads us to a definition.

**Definition 3** (Net Area).
Consider the region R bounded by the graph of a continuous function $f$ and the $x$-axis between $x = a$ and $x = b$. The *net area* of R is the sum of the areas of the parts of R that lie above the $x$-axis minus the sum of the areas of the parts of R that lie below the $x$-axis on $[a,b]$.

Riemann sums for $f$ on $[a,b]$ give approximations to the net area of the region bounded by the graph of $f$ and the $x$-axis between $x = a$ and $x = b$, where $a < b$. How can we make these approximations exact? Well, if $f$ is continuous on $[a,b]$, it is reasonable to expect the Riemann sum approximations to approach the exact value of the net area as the number of subintervals $n \to \infty$ and as the length of the subintervals $\Delta x \to 0$. In terms of limits, this would mean

$$\text{net area } = \lim_{n\to\infty} \sum_{k=1}^{n} f\left(x_k^*\right)\Delta x.$$

The Riemann sums we have used so far involve regular partitions in which the subintervals have the same length, $\Delta x$. We now introduce partitions of $[a,b]$ in which the lengths of the subintervals are not necessarily equal. A general partition of $[a,b]$ consists of the $n$ subintervals $[x_0,x_1]$, $[x_1,x_2]$, $\ldots$, $[x_{n-1},x_n]$ where $x_0 = a$ and $x_n = b$. The length of the $k^{th}$ subinterval is $\Delta x_k = x_k - x_{k-1}$, for $k = 1,\ldots,n$. We let $x_k^*$ be any point in the subinterval $[x_{k-1},x_k]$. This general partition is used to define the general Riemann sum.

**Definition 4** (General Riemann Sum).
Let $[x_0,x_1]$, $[x_1,x_2]$, $\ldots$, $[x_{n-1},x_n]$ be subintervals of $[a,b]$ with

$$a = x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n = b.$$

For $k = 1,\ldots,n$, let $\Delta x_k$ be the length of the subinterval $[x_{k-1},x_k]$ and let $x_k^*$ be any point in $[x_{k-1},x_k]$. Then, if $f$ is defined on $[a,b]$, the sum

$$\sum_{k=1}^{n} f\left(x_k^*\right)\Delta x_k = f\left(x_1^*\right)\Delta x_1 + f\left(x_2^*\right)\Delta x_2 + \ldots + f\left(x_n^*\right)\Delta x_n$$

is called a general Riemann sum for $f$ on $[a,b]$.

Now consider the limit of $\sum_{k=1}^{n} f\left(x_k^*\right)\Delta x$ as $n \to \infty$ and as all the $\Delta x_k \to 0$. We let $\Delta$ denote the largest value of $\Delta x_k$; that is, $\Delta = \max\left\{\Delta x_1, \Delta x_2, \ldots, \Delta x_n\right\}$. Observe that if $\Delta \to 0$, then $\Delta x_k \to 0$, for $k = 1, 2, \ldots, n$. For the limit $\lim_{\Delta \to 0} \sum_{k=1}^{n} f\left(x_k^*\right)\Delta x$ to exist, it must have the same value over all general partitions of $[a, b]$ and for all choices of $x_k^*$ on a partition.

**Definition 5** (Definite Integral).
A function $f$ defined on $[a, b]$ is integrable on $[a, b]$ if $\lim_{\Delta \to 0} \sum_{k=1}^{n} f\left(x_k^*\right)\Delta x$ exists and is unique over all partitions of $[a, b]$ and all choices of $x_k^*$ on a partition. This limit is the definite integral of $f$ from $a$ to $b$, which we write

$$\int_a^b f(x)\ dx = \lim_{\Delta \to 0} \sum_{k=1}^{n} f\left(x_k^*\right)\Delta x.$$

When this limit exists, it equals the net area of the region bounded by the graph of $f$ and the x-axis on $[a, b]$.

Let's look at a few useful theorems regarding definite integrals.

**Theorem 2** (Integrable Functions).
*If $f$ is continuous on $[a, b]$ or bounded on $[a, b]$ with a finite number of discontinuities, then $f$ is integrable on $[a, b]$.*

**Theorem 3** (Properties of Definite Integrals).
*Let $f$ and $g$ be integrable functions on an interval that contains $a$, $b$, and $p$.*

1. $\int_a^a f(x) dx = 0$
2. $\int_a^b f(x)\ dx = -\int_b^a f(x)\ dx$
3. $\int_a^b \big(f(x) + g(x)\big)\ dx = \int_a^b f(x)\ dx + \int_a^b g(x)\ dx$
4. $\int_a^b c f(x)\ dx = c\int_a^b f(x)\ dx$
5. $\int_a^b f(x)\ dx = \int_a^p f(x)\ dx + \int_p^b g(x)\ dx$
6. *The function $|f|$ is integrable on $[a, b]$, and $\int_a^b |f(x)|\ dx$ is the sum of the areas of the regions bounded by the graph of $f$ and the x-axis on $[a, b]$.*

**Example 3.**

Assume that $\int_0^5 f(x)\ dx = 3$ and $\int_0^7 f(x)\ dx = -10$. Evaluate the following integrals, if possible.

1. $\int_0^7 2f(x)\ dx$
2. $\int_5^7 f(x)\ dx$
3. $\int_5^0 f(x)\ dx$
4. $\int_7^0 6f(x)\ dx$
5. $\int_0^7 |f(x)|\ dx$

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1. $\int_0^7 2f(x)\ dx = 2\int_0^7 f(x)\ dx = 2(-10) = -20$
2. $\int_5^7 f(x)\ dx = \int_0^7 f(x)\ dx - \int_0^5 f(x)\ dx = -10 - 3 = -13$
3. $\int_5^0 f(x)\ dx = -\int_0^5 f(x)\ dx = -3$
4. $\int_7^0 6f(x)\ dx = -\int_0^7 6f(x)\ dx = -6\int_0^7 f(x)\ dx = -6(-10) = 60.$
5. $\int_0^7 |f(x)|\ dx$ cannot be evaluated without knowing the intervals on which $f$ is positive and negative. It could have any value greater than or equal to 10.

# 3

## Section 5.3 - Fundamental Theorem of Calculus

Evaluating definite integrals using limits of Riemann sums, as described in Section 5.2, is usually not possible or practical. Fortunately, there is a powerful and practical method for evaluating definite integrals, which is developed in this section. Along the way, we discover the inverse relationship between differentiation and integration, expressed in the most important result of calculus, the Fundamental Theorem of Calculus.

> **Definition 6** (Area Function).
> Let $f$ be a continuous function for $t \geq a$. The area function for $f$ with left endpoint $a$ is
>
> $$A(x) = \int_a^x f(t)\ dt,$$
>
> where $x \geq a$. The area function gives the net area of the region bounded by the graph of $f$ and the $t$-axis on the interval $[a, x]$.

Notice that $x$ is the upper limit of the integral and the independent variable of the area function: As $x$ changes, so does the net area under the curve. Because the symbol $x$ is already in use as the independent variable for A, we must choose another symbol for the variable of integration. Any symbol – except $x$ – can be used because it is a dummy variable; we have chosen $t$ as the integration variable.

Now, we may state the fundamental theorem of calculus. For intuition as to why this theorem is true, see Example 2 and the section titled "Fundamental Theorem of Calculus" in your textbook.

> **Theorem 4** (Fundamental Theorem of Calculus - Part I).
> *If $f$ is continuous on $[a, b]$, then the area function*
>
> $$A(x) = \int_a^x f(t)\ dt, \quad \text{for } a \leq x \leq b,$$
>
> *is continuous on $[a, b]$ and differentiable on $[a, b]$. The area function satisfies $A'(x) = f(x)$. Equivalently,*
>
> $$A'(x) = \frac{d}{dx}\int_a^x f(t)\ dt = f(x),$$
>
> *which means that the area function of $f$ is an antiderivative of $f$ on $[a, b]$.*

Given that A is an antiderivative of $f$ on $[a, b]$, it is one short step to a powerful method for evaluating definite integrals. Recall now that any two antiderivatives of $f$ differ by a constant. Assuming that F is any other antiderivative of $f$ on $[a, b]$, we have

$$F(x) = A(x) + C, \quad \text{for } a \leq x \leq b.$$

Noting that $A(a) = 0$, it follows that

$$F(b) - F(a) = \Big(A(b) + C\Big) - \Big(A(a) + C\Big) = A(b).$$

Writing $A(b)$ in terms of a definite integral leads to the remarkable result

$$A(b) = \int_a^b f(x)\ dx = F(b) - F(a).$$

**Theorem 5** (Fundamental Theorem of Calculus - Part II).
*If $f$ is continuous on $[a,b]$ and $F$ is any antiderivative of $f$ on $[a,b]$, then*

$$\int_a^b f(x)\ dx = F(b) - F(a).$$

It is customary and convenient to denote the difference $F(b) - F(a)$ by $F(x)\Big|_a^b$. Using this shorthand, the Fundamental Theorem is summarized as saying $\int_a^b f(x)\ dx = F(x)\Big|_a^b$. It is also worth pausing to observe that the two parts of the Fundamental Theorem express the inverse relationship between differentiation and integration. Part I of the Fundamental Theorem says

$$\frac{d}{dx}\int_a^x f(t)\ dt = f(x),$$

or, that is to say, that the derivative of the integral of $f$ is $f$ itself. Noting that $f$ is an antiderivative of $f'$, Part 2 of the Fundamental Theorem says

$$\int_a^b f'(x)\ dx = f(b) - f(a),$$

or the definite integral of the derivative of $f$ is given in terms of $f$ evaluated at two points. In other words, the integral "undoes" the derivative.

**Example 4.**
Evaluate the definite integral $\int_0^{10} \left(60x - 6x^2\right)\ dx$ using the Fundamental Theorem of Calculus.

Using the antiderivative rules of Section 4.9, an antiderivative of $60x - 6x^2$ is $30x^2 - 2x^3$. By the Fundamental Theorem, the value of the definite integral is

$$\begin{aligned}\int_0^{10} \left(60x - 6x^2\right)\ dx &= \left(30x^2 - 2x^3\right)\Big|_0^{10} \\ &= \left(30(10)^2 - 2(10)^3\right) - \left(30(0)^2 - 2(0)^3\right) \\ &= (3000 - 2000) - 0 \\ &= 1000.\end{aligned}$$

**Example 5.**

Evaluate the definite integral $\int_0^{2\pi} 3\sin(x)\ dx$ using the Fundamental Theorem of Calculus.

Using the antiderivative rules of Section 4.9, an antiderivative of $3\sin(x)$ is $-3cos(x)$. By the Fundamental Theorem, the value of the definite integral is

$$\begin{aligned}\int_0^{2\pi} 3\sin(x)\ dx &= -3cos(x)\Big|_0^{2\pi} \\ &= \Big(-3cos(2\pi)\Big) - \Big(-3cos(0)\Big) \\ &= -3-(-3) \\ &= 0.\end{aligned}$$

**Example 6.**

Evaluate the definite integral $\int_{1/16}^{1/4} \frac{\sqrt{t}-1}{t}\ dt$ using the Fundamental Theorem of Calculus.

First, we simplify the integrand

$$\frac{\sqrt{t}-1}{t} = \frac{1}{\sqrt{t}} - \frac{1}{t} = t^{-1/2} - t^{-1}.$$

Using the antiderivative rules of Section 4.9, an antiderivative of $t^{-1/2} - t^{-1}$ is $2t^{1/2} - ln|t|$. By the Fundamental Theorem, the value of the definite integral is

$$\begin{aligned}\int_{1/16}^{1/4} \frac{\sqrt{t}-1}{t}\ dt &= \Big(2t^{1/2} - ln|t|\Big)\Big|_{1/16}^{1/4} \\ &= \Big(2\sqrt{1/4} - ln(1/4)\Big) - \Big(2\sqrt{1/16} - ln(1/16)\Big) \\ &= 2(1/2) - ln(1/4) - 2(1/4) + ln(1/16) \\ &= 1 - ln(1/4) - (1/2) + ln(1/16) \\ &= 1/2 - ln(4) \\ &\approx -0.8863.\end{aligned}$$

**Example 7.** Use the Fundamental Theorem of Calculus to simplify the expression

$$\frac{d}{dx}\int_1^x sin^2(t)\ dt.$$

Using Part I of the Fundamental Theorem of Calculus, we see that

$$\frac{d}{dx}\int_1^x sin^2(t)\ dt = sin^2(x).$$

**Example 8.** Use the Fundamental Theorem of Calculus to simplify the expression

$$\frac{d}{dx}\int_x^5 \sqrt{t^2+1}\ dt.$$

To apply Part I of the Fundamental Theorem, the variable must appear in the upper limit. Therefore, we use the fact that

$$\int_x^5 \sqrt{t^2+1}\ dt = -\int_5^x \sqrt{t^2+1}\ dt$$

and then apply the Fundamental Theorem:

$$\frac{d}{dx}\int_x^5 \sqrt{t^2+1}\ dt = -\frac{d}{dx}\int_5^x \sqrt{t^2+1}\ dt = -\sqrt{x^2+1}.$$

**Example 9.** Use the Fundamental Theorem of Calculus to simplify the expression

$$\frac{d}{dx}\int_0^{x^2} cos\left(t^2\right)\ dt.$$

The upper limit of the integral is not $x$, but a function of $x$. Therefore, the function to be differentiated is a composite function, which requires the Chain Rule. We let $u = x^2$ to produce

$$y = g(u) = \int_0^u cos\left(t^2\right)\ dt.$$

So, by the Chain Rule,

$$\begin{aligned}\frac{d}{dx}\int_0^{x^2} cos\left(t^2\right)\ dt &= \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}\\ &= \left(\frac{d}{du}\int_0^u cos\left(t^2\right)\ dt\right)\left(2x\right)\\ &= \left(cos\left(u^2\right)\right)\left(2x\right)\\ &= 2xcos\left(x^4\right)\end{aligned}$$

## Section 5.4 - Working with Integrals

With the Fundamental Theorem of Calculus in hand, we may begin an investigation of integration and its applications. In this section, we discuss the role of symmetry in integrals, we use the slice-and-sum strategy to define the average value of a function, and we explore a theoretical result called the Mean Value Theorem for Integrals.

**Theorem 6** (Integrals of Even and Odd Functions).
*Let $a$ be a positive real number and let $f$ be an integrable function on the interval $[-a,a]$.*

- *If $f$ is even (i.e. if $f(-x) = f(x)$), then* $\int_{-a}^{a} f(x)\ dx = 2\int_{0}^{a} f(x)\ dx.$
- *If $f$ is odd (i.e. if $f(-x) = -f(x)$), then* $\int_{-a}^{a} f(x)\ dx = 0.$

**Example 10.** Evaluate the integral $\int_{-2}^{2} \left(x^4 - 3x^3\right)\ dx.$

Note that $x^4 - 3x^3$ is neither odd nor even so Theorem 6 cannot be applied directly. However, as $x^4$ is even and $x^3$ is odd, we can split the integral and then use symmetry:

$$\begin{aligned}
\int_{-2}^{2} \left(x^4 - 3x^3\right)\ dx &= \int_{-2}^{2} x^4\ dx - 3\int_{-2}^{2} x^3\ dx \\
&= 2\int_{0}^{2} x^4\ dx - 0 \\
&= 2\left(\frac{x^5}{5}\right)\Big|_0^2 \\
&= 2\left(\frac{2^5}{5} - \frac{0^5}{5}\right) = \frac{64}{5}.
\end{aligned}$$

**Example 11.** Evaluate the integral $\int_{-\pi/2}^{\pi/2} \left(cos(x) - 4sin^3(x)\right)\ dx.$

The $cos(x)$ term is an even function, so it can be integrated on the interval $[0,\pi/2]$. What about $sin^3(x)$? It is an odd function raised to an odd power, which results in an odd function; its integral on $[-\pi/2,\pi/2]$ is zero. Therefore,

$$\begin{aligned}
\int_{-\pi/2}^{\pi/2} \left(cos(x) - 4sin^3(x)\right)\ dx &= 2\int_{0}^{\pi/2} cos(x)\ dx - 4\int_{-\pi/2}^{\pi/2} sin^3(x)\ dx \\
&= 2\sin(x)\Big|_0^{\pi/2} - 0 \\
&= 2(1-0) = 2.
\end{aligned}$$

As the definite integral $\int_a^b f(x)\ dx$ measures the net area of the region bounded by the graph of $f$ and the $x$-axis on the interval $[a,b]$, notice that there exists a rectangle with witdth $b-a$ (i.e. the lenght of the interval) and height $h$ such that Area of rectangle $= h(b-a) = \int_a^b f(x)\ dx$. Namely, this height is just $h = \frac{1}{b-a}\int_a^b f(x)\ dx$. We call this the *average value* of $f$ on the interval $[a,b]$.

**Definition 7** (Average Value of a Function).
The average value of an integrable function $f$ on the interval $[a,b]$ is

$$\overline{f} = \frac{1}{b-a}\int_a^b f(x)\ dx.$$

**Example 12.** A hiking trail has an elevation given by

$$f(x) = 60x^3 - 650x^2 + 1200x + 4500,$$

where $f$ is measured in feet above sea level and $x$ represents horizontal distance along the trail in miles, with $0 \le x \le 5$. What is the average elevation of the trail?

If one graphs the function, we may notice that the trail ranges between elevations of about 2000 and 5000 ft. If we let the endpoints of the trail correspond to the horizontal distances $a = 0$ and $b = 5$, the average elevation of the trail in feet is

$$\begin{aligned}\overline{f} &= \frac{1}{b-a}\int_a^b f(x)\ dx \\ &= \frac{1}{5-0}\int_0^5 \left(60x^3 - 650x^2 + 1200x + 4500\right)\ dx \\ &= \frac{1}{5}\left(\frac{60}{4}x^4 - \frac{650}{3}x^3 + \frac{1200}{2}x^2 + 4500x\right)\Bigg|_0^5 \\ &= \frac{11875}{3} \approx 3958.33\end{aligned}$$

So, the average elevation of the trail is slightly less than 3960 ft.

The average value of a function brings us close to an important theoretical result. The Mean Value Theorem for Integrals says that if $f$ is continuous on $[a,b]$, then there is at least one point $c$ in the interval $[a,b]$ such that $f(c)$ equals the average value of $f$ on $[a,b]$. In other words, the horizontal line $y = \overline{f}$ intersects the graph of $f$ for some point $c$ in $[a,b]$. Here, continuity is key – if $f$ were not continuous, such a point might not exist.

**Theorem 7** (Mean Value Theorem for Integrals).
*Let $f$ be continuous on the interval $[a,b]$. There exists a point $c$ in $[a,b]$ such that*

$$f(c) = \overline{f} = \frac{1}{b-a}\int_a^b f(x)\ dx.$$

**Example 13.** Find the point(s) on the interval $[0,1]$ at which $f(x) = 2x(1-x)$ equals its average value on $[0,1]$.

The average value of $f$ on $[0,1]$ is

$$\overline{f} = \frac{1}{1-0}\int_0^1 2x(1-x)\ dx = \left(x^2 - \frac{2}{3}x^3\right)\Big|_0^1 = \frac{1}{3}.$$

Now, we must find the points on $[0,1]$ at which $f(x) = \frac{1}{3}$. Notice, this is equivalent to solving $f(x) - \frac{1}{3} = 0$, i.e. solving the equation $2x(1-x) - \frac{1}{3} = 0$. Using the quadratic formula, the two solutions are

$$x = \frac{1-\sqrt{1/3}}{2} \approx 0.211 \quad \text{and} \quad x = \frac{1+\sqrt{1/3}}{2} \approx 0.789.$$

So, for these values of $x$, we may conclude that $f$ equals its average value on $[0,1]$.

# 5

## Section 6.1 - Velocity and net Change

In previous chapters, we established the relationship between the position and velocity of an object moving along a line. With integration, we can now say much more about this relationship. Once we relate velocity and position through integration, we can make analogous observations about a variety of other practical problems, which include fluid flow, population growth, manufacturing costs, and production and consumption of natural resources. The ideas in this section come directly from the Fundamental Theorem of Calculus, and they are among the most powerful applications of calculus.

**Definition 8** (Position, Velocity, Displacement, and Distance).

1. The position of an object moving along a line at time $t$, denoted $s(t)$, is the location of the object relative to the origin.
2. The velocity of an object at time $t$ is $v(t) = s'(t)$.
3. The displacement of the object between $t = a$ and $t = b > a$ is $s(b) - s(a) = \int_a^b v(t)\ dt$.
4. The distance traveled by the object between $t = a$ and $t = b > a$ is $\int_a^b |v(t)|\ dt$, where $|v(t)|$ is the speed of the object at time $t$.

Suppose we know the velocity of an object and its initial position $s(0)$. Suppose our goal is to find the position $s(t)$ of the object at some future time $t \geq 0$. Conveniently, the Fundamental Theorem of Calculus gives us the answer directly in a fashion we summarize with the following theorem.

**Theorem 8** (Position from Velocity).
*Given the velocity $v(t)$ of an object moving along a line and its initial position $s(0)$, the position function of the object for future times $t \geq 0$ is*

$$s(t) = s(0) + \int_0^t v(x)\ dx.$$

A similar trick works for velocity and acceleration as well.

**Theorem 9** (Velocity from acceleration).
*Given the acceleration $a(t)$ of an object moving along a line and its initial velocity $s(0)$, the velocity function of the object for future times $t \geq 0$ is*

$$v(t) = v(0) + \int_0^t a(x)\ dx.$$

**Example 14.** A jogger runs along a straight road with velocity (in mi/hr) $v(t) = 2t^2 - 8t + 6$, for $0 \le t \le 3$, where $t$ is measured in hours.

1. Determine when the jogger moves in the positive direction and when she moves in the negative direction.
2. Find the displacement of the jogger (in miles) on the time intervals $[0,1]$, $[1,3]$, and $[0,3]$. Interpret your result.
3. Find the distance traveled over the interval $[0,3]$.

1. By solving $v(t) = 0$, we may find when the sign of $v(t)$ changes. Factoring and solving,

$$v(t) = 2t^2 - 8t + 6 = 2(t-1)(t-3) = 0,$$

we find that the velocity is zero at $t = 1$ and $t = 3$. These values are the $t$-intercepts of the graph of $v$, which is an upward-opening parabola with a $v$-intercept of 6. The velocity is positive on the interval $0 \le t < 1$, which means the jogger moves in the positive $s$ direction. For $1 < t < 3$, the velocity is negative and the jogger moves in the negative $s$ direction.

2. To find the displacement of the jogger (in miles) on the time interval $[0,1]$, we calculate

$$\int_0^1 v(t)\ dt = \int_0^1 \left(2t^2 - 8t + 6\right)\ dt = \left(\frac{2}{3}t^3 - 4t^2 + 6t\right)\Big|_0^1 = \frac{8}{3}.$$

Similarly, over the interval $[1,3]$, we calculate

$$\int_1^3 v(t)\ dt = \int_1^3 \left(2t^2 - 8t + 6\right)\ dt = \left(\frac{2}{3}t^3 - 4t^2 + 6t\right)\Big|_1^3 = -\frac{8}{3}.$$

Now, note that that the displacement of the jogger (in miles) over the interval $[0,3]$ is equal to the sum of the displacements over the intervals $[0,1]$ and $[1,3]$. So, over the interval $[0,3]$, the displacement is $(8/3) + (-8/3) = 0$, which means the jogger returns to the starting point after three hours.

3. To find the distance traveled over the interval $[0,3]$, we use information regarding the sign of $v(t)$ found in part 1 and calculate

$$\begin{aligned}\int_0^3 |v(t)|\ dt &= \int_0^1 \left(2t^2 - 8t + 6\right)\ dt + \int_1^3 \left(-\left(2t^2 - 8t + 6\right)\right)\ dt \\ &= \left(\frac{2}{3}t^3 - 4t^2 + 6t\right)\Big|_0^1 + \left(-\frac{2}{3}t^3 + 4t^2 - 6t\right)\Big|_1^3 \\ &= \frac{16}{3}.\end{aligned}$$

Alternatively, From part 2, we can deduce the total distance traveled by the jogger. On the interval $[0,1]$, the distance traveled is 8/3 mi; on the interval $[1,3]$, the distance traveled is also 8/3 mi. Therefore, the distance traveled on $[0,3]$ is 16/3 mi.

**Example 15.** A block hangs at rest from a massless spring at the origin ($s = 0$). At $t = 0$, the block is pulled downward 1/4 m to its initial position $s(0) = -1/4$ and released. Its velocity (in m/s) is given by $v(t) = 1/4\sin(t)$ for $t \geq 0$. Assume that the upward direction is positive. Find the position of the block for $t \geq 0$.

We let the positive direction be upward with the origin ($s = 0$) corresponding to the position where block hangs when at rest. The initial position of the block is $s(0) = -1/4$ m. The velocity (in m/s) is given by $v(t) = 1/4\sin(t)$ m/s. Integrating the velocity, the position is

$$\begin{aligned}
s(t) &= s(0) + \int_0^t v(x)\ dx \\
&= -\frac{1}{4} + \int_0^t \frac{1}{4}\sin(x)\ dx \\
&= -\frac{1}{4}\left(\frac{1}{4}cos(x)\right)\Big|_0^t \\
&= -\frac{1}{4} - \frac{1}{4}\Big(cos(t) - 1\Big) \\
&= -\frac{1}{4}cos(t).
\end{aligned}$$

**Example 16.** An artillery shell is fired directly upward with an initial velocity of 300 m/s from a point 30 m above the ground. Assume that only the force of gravity acts on the shell and it produces an acceleration of 9.8 m/s$^2$. Find the velocity of the shell while it is in the air.

We let the positive direction be upward with the origin ($s = 0$) corresponding to the ground. The initial velocity of the shell is $v(0) = 300$ m/s. The acceleration due to gravity is downward; therefore, $a(t) = -9.8$ m/s$^2$. Integrating the acceleration, the velocity is

$$v(t) = v(0) + \int_0^t a(x)\ dx = 300 + \int_0^t -9.8\ dx = 300 - 9.8t.$$

**Theorem 10** (Net Change and Future Value).
*Suppose a quantity* Q *changes over time at a known rate* Q′*. Then, the net change in* Q *between* $t = a$ *and* $t = b > a$ *is*

$$Q(b) - Q(a) = \int_a^b Q'(t)\ dt.$$

*Given the initial value* Q(0)*, the future value of* Q *at time* $t \geq 0$ *is*

$$Q(t) = Q(0) + \int_0^t Q'(x)\ dx.$$

**Example 17.** A culture of cells in a lab has a population of 100 cells when nutrients are added at time $t = 0$. Suppose the population $N(t)$ (in cells/hr) increases at a rate given by $N'(t) = 90e^{-0.1t}$. Find $N(t)$, for $t \geq 0$.

If one graphs the function $N'(t)$, it is clear that the growth rate is large when $t$ is small (plenty of food and space) and decreases as $t$ increases. Knowing that the initial population is $N(0) = 100$ cells, we can find the population $N(t)$ at any future time $t \geq 0$ as follows:

$$\begin{aligned} N(t) &= N(0) + \int_0^t N'(t)\ dt \\ &= 100 + \int_0^t 90e^{-0.1x}\ dx \\ &= 100 + \left(\frac{-90}{0.1}e^{-0.1x}\right)\Big|_0^t \\ &= 100 - 900e^{-0.1t} + 900 \\ &= 1000 - 900e^{-0.1t} \end{aligned}$$

Note that, if one graphs the population function, it is easy to see that the population increases, but at a decreasing rate. Note also that the population size approaches 1000 cells as $t \to \infty$, as

$$\lim_{t\to\infty} N(t) = \lim_{t\to\infty}\left(1000 - 900e^{-0.1t}\right) = 1000 - 0 = 1000.$$